i. Research Topic Using IEEE Terminology

- **Primary IEEE Preferred Term:** Anomaly Detection
- **Supporting IEEE Terms:** Temporal Graph Networks, Graph Neural Networks, Intrusion Detection, Network Security, Swarm Intelligence, Optimization, Dynamic Graphs, Log Analysis, and Class Imbalance.
- **Research Domain:** Computer Science, Artificial Intelligence, Cybersecurity, and Data Science.

ii. Research Paper Title

- **Paper Title :** Anomaly Detection From Log Files Using Temporal Graph Network
- **Keywords:** Temporal Graph Networks (TGN), Log Anomaly Detection, Particle Swarm Optimization (PSO), Continuous-Time Dynamic Graphs (CTDG), Network Intrusion Detection, Adaptive Thresholding, Class Imbalance, LANL Cyber Security Dataset.

iii. Control Variable

To ensure that any performance variation is caused solely by the threshold calibration strategy, all core experimental settings are kept constant throughout the study.

The control variables include:

- LANL Comprehensive Cyber Security Dataset (auth.txt.gz, flows.txt.gz, and redteam.txt.gz)
- Fixed chronological data partitioning (50% training, 25% validation, 25% testing)
- Eight-dimensional edge feature representation
- TGNMemory architecture with memory_dim = 100 and time_dim = 100
- IdentityMessage and LastAggregator modules
- LinkPredictor neural network architecture
- Learning rate = 0.0001
- Batch size = 2000
- Training epochs = 5
- Weighted BCEWithLogitsLoss function

iv. Independent Variable

PSO-Based Threshold Calibration: The independent variable in this study is the threshold calibration mechanism applied to the output scores of the Temporal Graph Network (TGN). A Particle Swarm Optimization (PSO) algorithm is employed to automatically determine the optimal anomaly detection threshold by balancing detection recall and false-positive rates on the validation dataset.

v. Dependent Variable

The dependent variables are the anomaly detection performance metrics measured on the unseen test dataset:

- Precision
- Recall (Sensitivity)
- F1-Score
- False Positive Rate (FPR)
- Area Under the Receiver Operating Characteristic Curve (AUROC)

These metrics are used to evaluate the effectiveness of each threshold calibration strategy.

vi. Research Intervention

This study introduces a Particle Swarm Optimization (PSO)-based threshold calibration mechanism into a Temporal Graph Network (TGN) framework for network log anomaly detection. The intervention aims to improve anomaly detection performance under highly imbalanced cybersecurity data by introducing an adaptive PSO-based threshold optimization mechanism into a Temporal Graph Network framework.

The research procedure consists of five phases:

- **Phase 1 – Data Preprocessing**

  Authentication logs are processed, labeled using Red Team attack records, and transformed into structured features. Normal events are undersampled while all attack events are retained.

- **Phase 2 – Dynamic Graph Construction**

  Authentication and network flow events are converted into a Continuous-Time Dynamic Graph (CTDG), preserving the temporal relationships between entities.

- **Phase 3 – Tensor Encoding**

  Graph interactions are transformed into PyTorch tensors using an eight-dimensional edge feature representation.

- **Phase 4 – Temporal Graph Network Training**

  A Temporal Graph Network (TGN) model is trained using the first 50% of chronological events. The model learns temporal interaction patterns among users, hosts, and network entities.

- **Phase 5 – PSO-Based Threshold Optimization and Evaluation**

  Particle Swarm Optimization is applied to the validation set to automatically discover the optimal anomaly decision threshold. The optimized threshold is then evaluated on an unseen test set to measure detection performance.

The proposed intervention enables adaptive threshold selection, improves anomaly recall, and reduces false-positive alerts in highly imbalanced network security environments.